Chapter 4: Familiarization with Construction Equipment

4.1 Advantages and Drawbacks of Using Equipment

Advantages:

- Increase the rate of output of work
- Reduce the overall construction cost
- Carry out activities which can’t be done by manually
- Equipment can work in adverse weather climate and topography
- Reduce fatigue and eliminates hazards & health issues
- Maintain the planned rate of production where there is shortage of labor
- Maintain high quality standards
- Equipment is free from social and emotional factors
- It helps maintaining construction site safer and cleaner

Advantages and Disadvantages Using Equipment

Drawbacks:

- If the machine breaks down, it can cause delay of construction
- Electric tools can cause short circuit which may result in fire
- Only trained and skilled labor can operate the equipment
- It may harm health of the people on site
- Heavy investment for buying equipment
- Large inventory of spare parts
- May create social problem due the unemployment of people
- Reselling / disposing off construction equipment will be a problem

4.2 Equipment Used in Different Types of Construction

A) Earth Work Equipment

➢ Excavating Equipment:

Excavator, Back Hoes, Dozer, Scraper, Drag Line, Power Sovel & Grader

➢ Watering and Dewatering Equipment:

- Displacement Pump (Reciprocating Pump, Diaphram)
- Centrifugal Pump (Conventional, Self-Priming, Air-Operated)

➢ Loading Equipment:

➢ Transporting Equipment:

Dozer, Grader, Scraper, Power Sovel, Drag Line, Tractor, Triger, Dumper

➢ Compacting Equipment:

Tamping Rod, Hand Compactor, Tamping Roller, Smooth Wheel Roller, Pneumatic Roller, Sheep Foot Roller, Vibrating Roller, Rammer, Frog Rammer,

Equipment Used in Different Types of Construction

- **Drilling and Blasting Equipment:**

Jack Hammer, Drifter, Wagon Drill, Track Mounted Drill,

B) Lifting and Erecting Equipment: Fixed Crane (Tower Crane)

C) Concerting Equipment:

- **Aggregate Production Equipment:**

Primary Crusher, Secondary Crusher, Tertiary Crusher, Screening

- **Aggregate Handling Equipment:**

Truck, Dumper, Conveyor Belt

- **Concrete Mixing Equipment**:

Hand fed drum mixture, Self loading drum mixture, Concrete mixing pan, Transit mixture, Mixing plant

Equipment Used in Different Types of Construction

- **Concrete Transportation (Handling & Placing):**

Metal pan, Wheel barrow, Minidumper, Crane, Conveyor belt system, Concrete pump,

- **Concrete Compaction Equipment:**

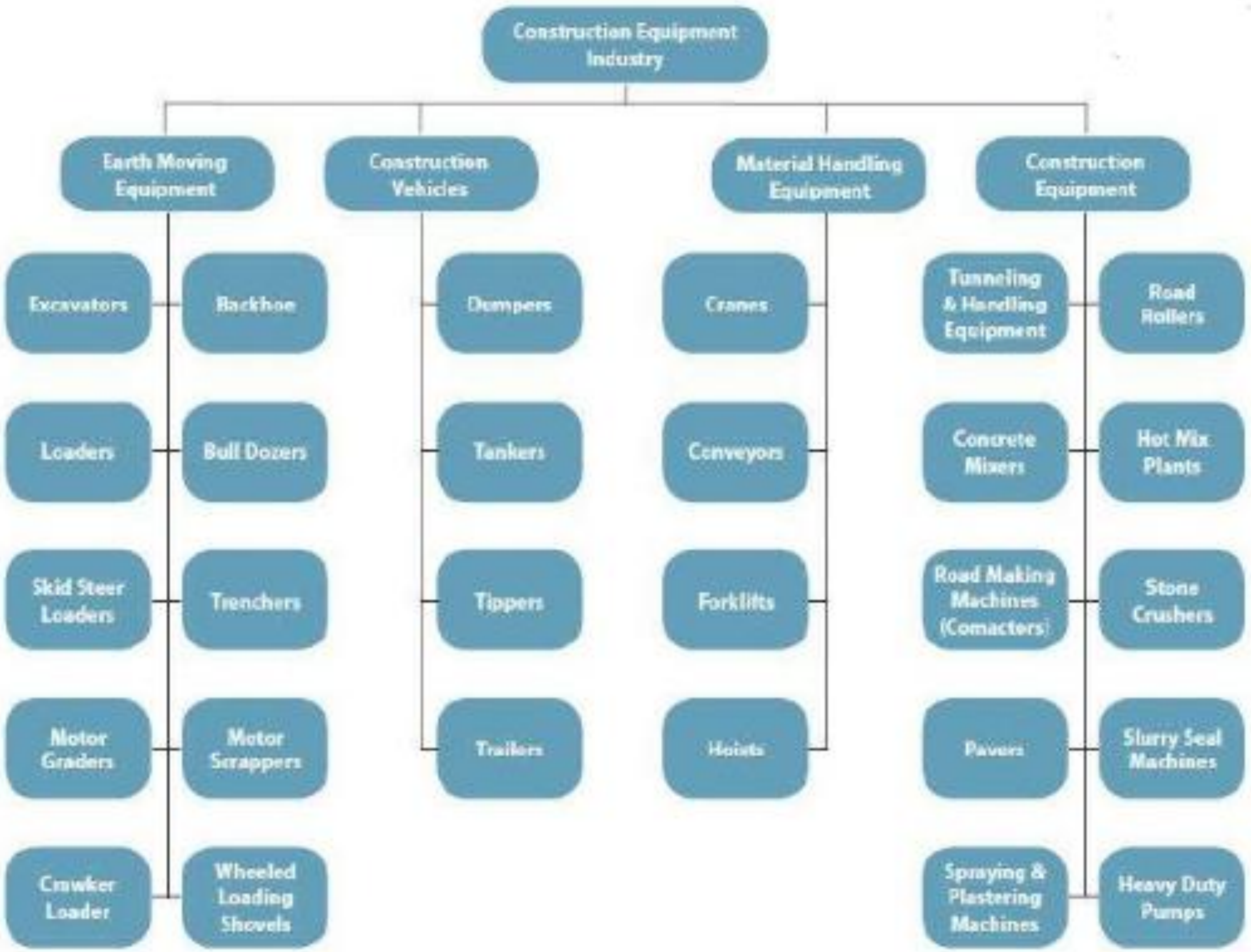
Needle vibrator, Surface vibrator, Form vibrator, Tamping rod

D) Tunneling Equipment:

Tunnel boring machine, Drilling machine, Electric detonating arrangement, Ventilating arrangement, Wheel barrow, Compressor, Drilling machine

E) Foundation Equipment: Pile boring machine, Pile driving machine

F) Steel Construction Equipment:


Construction Equipment Industry
Earth Moving Equipment
Construction Vehicles
Material Handling Equipment
Construction Equipment
Excavators
Backhoe
Loaders
Bull Dozers
Skid Steer Loaders
Trenchers
Motor Graders
Motor Scrappers
Crawker Loader
Wheeled Loading Shovels
Dumpers
Tankers
Tippers
Trailers
Cranes
Conveyors
Forklifts
Hoists
Tunneling & Handling Equipment
Road Rollers
Concrete Mixers
Hot Mix Plants
Road Making Machines (Comacters)
Stone Crushers
Pavers
Slurry Seal Machines
Spraying & Plastering Machines
Heavy Duty Pumps

Equipment Used in Different Types of Construction

Equipment for Excavation:

Excavator

- Excavators are important and widely used equipment in construction industry.
- They use for excavation as well as other many purposes like heavy lifting, demolition, river dredging, cutting of trees etc.
- Excavators contains a long arm and a cabinet.
- At the end of long arm digging bucket is provided.
- Cabinet is the place provided for machine operator.
- This whole cabin arrangement can be rotatable up to 360 degree.
- Excavators are available in both wheeled and tracked forms of vehicles.

Excavator

Tracked Excavator

Excavator

Levelled Diagram

Backhoe

- Backhoe is another widely used equipment which is suitable for multiple purposes.
- The name itself telling that the hoe arrangement is provided on the back side of vehicle.
- While loading bucket is provided in the front.
- This is well useful for excavating trenches below the machine level
- Using front bucket loading, unloading and lifting of materials can be done.
- The basic parts are boom, Jack boom, boom foot drum, boom sheave, stick sheave, bucket and bucket sheave.

Backhoe

Backhoe

Backhoe

① Bucket
② Arm
③ Hydraulic cylinder
④ Boom
⑤ Cab
⑥ Track frame
⑦ Engine
⑧ Counterweight
1
2
3
4
5
6
7
8

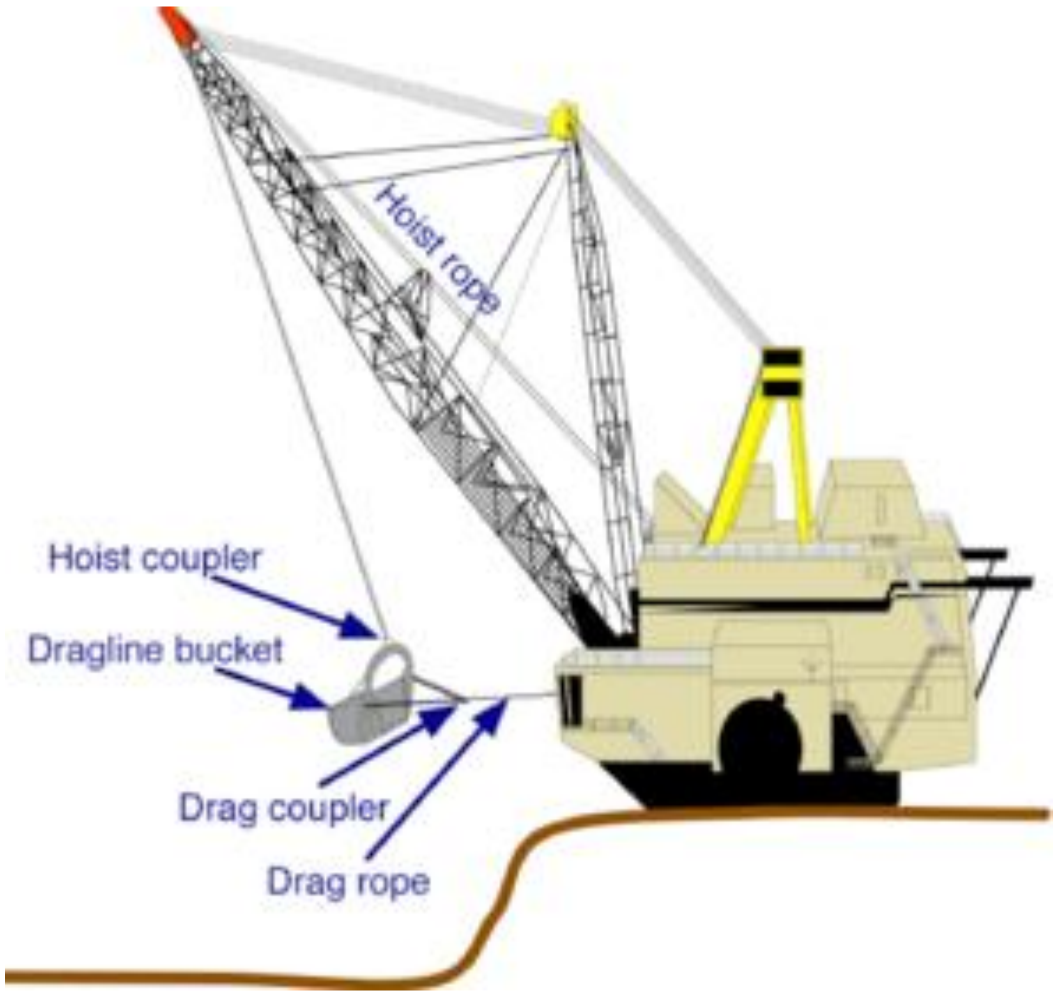
Drag Line Excavator

- Dragline excavator is another heavy equipment used in construction generally used for larger depth excavations.
- It consists a long length boom and digging bucket is suspended from the top of the boom using cable.
- It is used for the construction of ports, for excavations under water, sediment removal in water bodies

Drag line Excavator

Drag line Excavator

Drag Line Excavator



Bulldozer

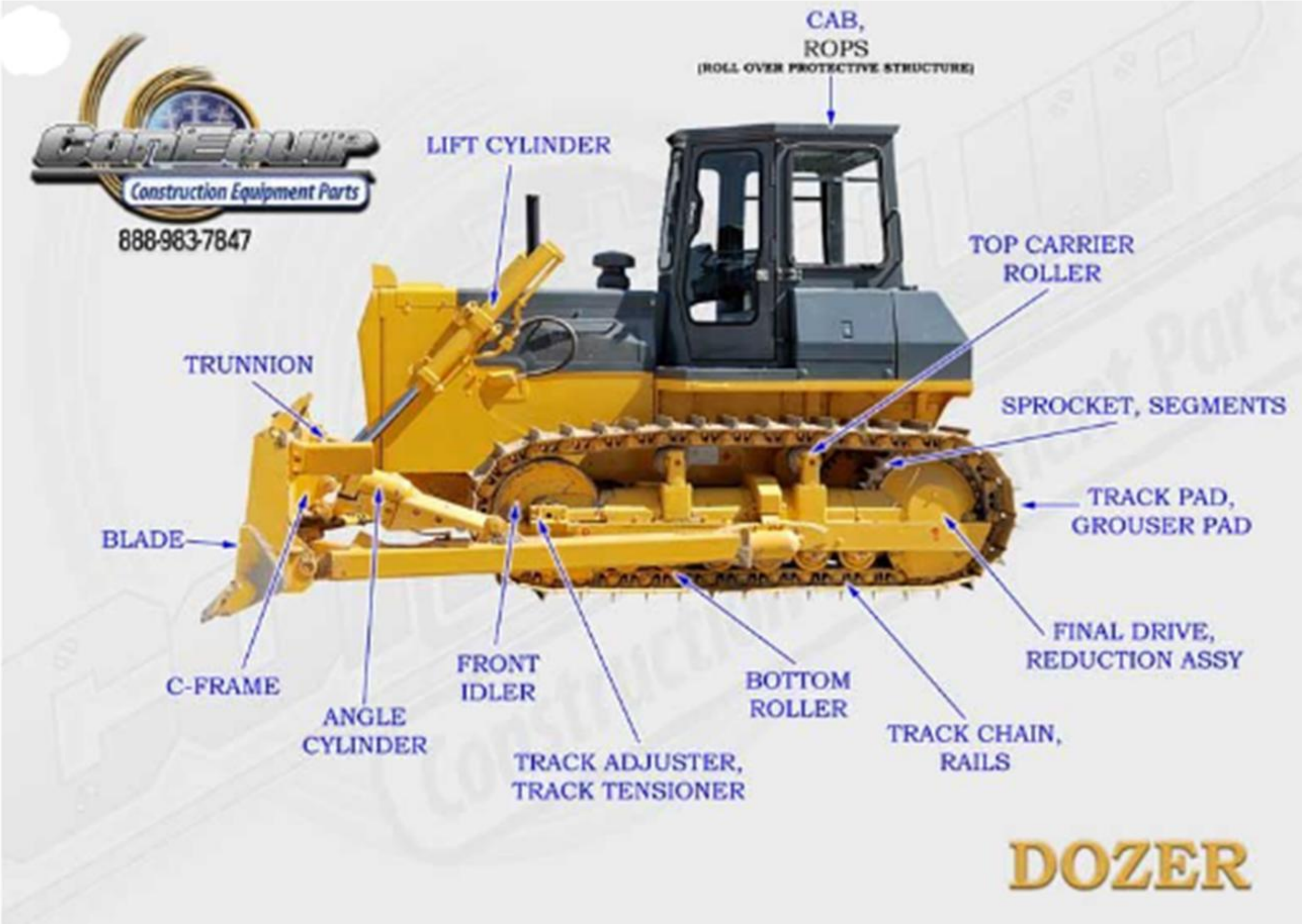
- Bulldozer is soil excavating equipment used to remove the topsoil, rock strata, lifting of soil.
- The removal of soil is done by the sharp edged wide metal plate provided at its front.
- This plate can be lowered and raised using hydraulic pistons.
- A bulldozer is a crawler equipped with a substantial metal plate and at the rear with a claw like device to loosen densely compacted materials
- It is used to push large quantities of soil, sand, rubble etc.

Bulldozer

Bulldozer

Bulldozer

Levelled Diagram:



Bulldozer

① Ripper
② Final Drive
③ Cab
④ Tracks/Tires
⑤ Engine
⑥ Push Frame
⑦ Blade

Grader

- It is used in road construction
- It is mainly used to level the soil surface.
- It contains a horizontal blade in between front and rear wheels
- This blade is lowered in to the ground while working.
- Operating cabin is provided on the top of rear axle arrangement.
- Motor Graders are also used to remove snow or dirt from the roads
- It is used to flatten the surface of soil before laying asphalt layer.
- It is also used to remove unnecessary soil layer from the ground.

Grader

Graders are classified as below

a) Based on weight:

- ➢Heavy (17-23 tons)
- ➢Medium (10-12 tons)
- ➢Light (＜10 tons)

b) Based on design of running gears, as four wheelers & six-wheelers

c) Based on control system, as mechanical and hydraulic grader

Grader

Grader

Wheel Tractor Scraper

- It is earth moving equipment used to provide flatten soil surface through scrapping.
- Front part contains wheeled tractor vehicle and rear part contains a scrapping arrangement such as horizontal front blade, conveyor belt and soil collecting hopper.
- When the front blade is lowered onto the ground and vehicle is moved, the blade starts digging the soil above the blade level and the soil excavated is collected in hopper through conveyor belt.
- When the hopper is full, the rear part is raised from the ground and hopper is unloaded at soil dump yard.

Wheel Tractor Scraper

Scraper

Trencher

- Trencher or Trenching machine is used to excavate trench in soil.
- Trench is generally used for pipeline laying, cable laying, drainage purposes etc.
- Trenching machine is available in two types namely chain trencher and wheeled trencher.
- A trencher contains a fixed long arm around which digging chain is provided.
- Wheeled trencher contains a metal wheel with digging tooth around it which excavates hard soil layer.
- Both types of trenchers are available in tracked as well as wheeled vehicle forms.

Wheeled Trencher

Trencher

Loader

- Loader is used in construction site to load the material onto dumpers, trucks etc.
- The materials may be excavated soil, demolition waste, raw materials.
- A loader contain large sized bucket at its front with shorter moving arm.
- Loader may be either tracked or wheeled.
- Wheeled loader is widely used in sites while tracked or crawled loader is used in site where wheeled vehicle cannot reach.

Loader

Loader

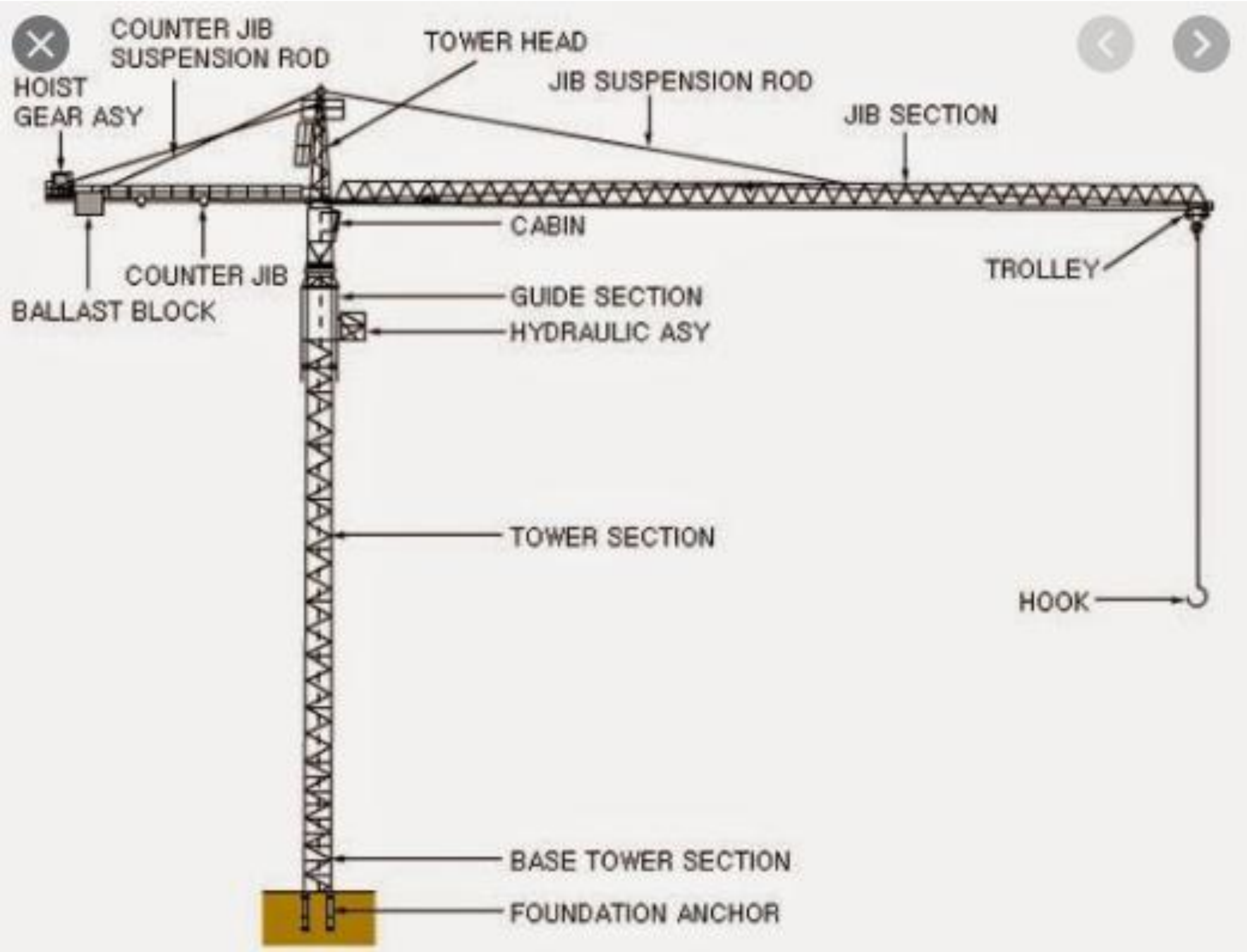
Lifting and Erecting Equipment (Tower Crane)

- Tower crane is fixed crane which is used for hoisting purposes in construction of tall structures.
- Heavy materials like pre-stressed concrete blocks, steel trusses. frames etc. can be easily lifted to required height using this type of equipment.
- It consists mast which is the vertical supporting tower.
- Jib which is operating arm of crane.
- Counter jib which is the other arm carries counter weight on rear side of crane.
- An operator cabin from which the crane can be operated.
- Tower crane is fabricated by the truss elements
- Movable (transporting) crane is also used in construction

Tower Crane

Tower Crane

Levelled Diagram



Paver

- Paver or Asphalt paver is pavement laying equipment.
- It is used in road construction.
- Paver contains a feeding bucket in which asphalt is continuously loaded by the dump truck.
- Paver distributes the asphalt evenly on the road surface with slight compaction.
- However a roller is required after laying asphalt layer for perfect compaction.

Asphalt Paver

Paver

Compactor

- Compactor or Roller is used to compact the material or earth surface.
- Different types of compactors are available for different **compacting** purposes.
- Smooth wheel rollers are used for compacting shallow layers of soil or asphalt etc.
- Sheep-foot roller is used for deep compaction purposes.
- Pneumatic tired roller is used for compacting fine grained soils, asphalt layers etc.

Smooth wheel Compactor

Compactor

Telehandler

- Telehandler is hoisting equipment used in construction to lift heavy materials up to required height
- To provide construction platform for workers at greater heights etc.
- It contains a long telescopic boom which can be raised or lowered or forwarded.
- Different types of arrangements like forklifts, buckets, cabin, lifting jibs etc. can be attached to the end of telescopic boom based on the requirement of job.

Telehandler

Telehandler

Feller Buncher

- Feller buncher is tree cutting heavy equipment.
- It is used to remove large trees in the construction field.
- It cuts the tree and grab it without felling.
- Likewise gathers all the cut down trees at one place which makes job easier for loaders and dump trucks.

Feller Buncher

Transporting Equipment

- ➢Trucks, wagons, and conveyor belts are hauling equipment.
- ➢They transport deposits from one place to another place.

Classification of truck:

- Size of engine
- No of gears
- Kind of drives
- No of wheels arrangement
- Method of dumping loads: rear/side
- Class of material hauled: earth /rock/coal /ore etc.
- Capacity in tones

Transporting & Compacting Equipment

Transporting Equipment:

Following also are fully or partially serves as the transporting equipment are: **Dozers, Grader, Scraper, Power Shovel, Drag Line, Dumper & Trippers.**

Compacting Equipment:

To increase the strength of the soil, compaction is done for which various types of equipment are available.

- Tamping rod
- Hand Compactor
- Tamping rod
- Pneumatic Roller
- Sheep Foot Roller
- Vibrating Roller

Compacting Equipment

Rammer:

- Simplest type of compacting equipment
- Heavy Iron weight is provided by a wooden handle.
- The soil surface can be compacted by lifting and releasing in a cycle.

Frog Rammer:

- Rammer with small ramming plate
- Jumps with the help of engine mounted on the top
- Suitable for very small trenches

Compactor:

- Hand held vibrating plate
- Diesel / Petrol engine mounted on springs over vibrating plate gives vibration

Compacting Equipment

Hand Roller/ Tamping Roller:

- Suitable for compacting small trenches (90 cm to 1.2m) wide
- Both vibrating and static load type available
- Single steel drum is the common type

Smooth wheel Roller

Pneumatic Roller

Sheep foot Roller

Vibrating Roller

Vibrating Roller

Dump Truck

- Dump truck is used in construction sites to carry the material in larger quantities from one site to another site or to the dump yard.
- Generally, in big construction site, off-road dump truck is used.
- These off-road dump truck contains large wheels with huge space for materials
- It enables to carry huge quantity of material in any type of ground conditions.

Dump Truck

Dump Truck

Dump Truck

Dump Truck

Pile Boring Equipment

- **Pile boring equipment is used to make bore holes in the construction site to install precast piles.**

Pile Driving Equipment

- It is also the heavy equipment used in construction site
- It is used in pile foundation construction.
- This equipment lifts the pile and holds it in proper position and drives into the ground up to required depth.
- Different types of pile driving equipment are available namely, piling rigs, piling hammer, hammer guides etc.
- In any case the pile is driven into the ground by hammering the pile top which is done hydraulically or by dropping.

Hydraulic Pile Driving Machine

Pile Driving Equipment

Power Shovel

Power Shovel

OBJECTIVE: For digging and loading earth or fragmented rock and for mineral extraction

Power Shovel

Power Shovel

- To excavate the earth and to load the trucks
- capable of excavating all types of earth except hard rock
- size varies from $0.5m^3$ to $5m^3$.
- Basics parts of power shovel including the track system, cabin, cables, rack, stick, boom foot-pin, saddle block, boom, boom point sheaves and bucket.

Bulldozer

Bulldozer

OBJECTIVE: To push large quantities of soil, sand, rubble, or any other such material

A bulldozer is a crawler (continuous tracked tractor) equipped with a substantial metal plate (known as a blade) and at the rear with a claw-like device (known as a ripper) to loosen densely compacted materials

Scrapers

Scrapper

OBJECTIVE: To remove scraps/dirt efficiently

Dragline

Dragline

- The drag line is so name because of its prominent operation of dragging the bucket against the material to be dug.
- Unlike the shovel, it has a long light crane boom and the bucket is loosely attached to the boom through cables.
- Because of this construction, a dragline can dig and dump over larger distances than a shovel can do.
- Drag lines are useful for digging below its track level and handling softer materials.

Equipment for Transportation and Compaction

Transportation & Compaction Equipment

Equipment for Transportation and Compaction

Compaction Equipment

Aggregate Production and Handling

Stone Crushing Machine

Aggregate Production and Handling

Aggregate Crushing plant:

Aggregate Production and Handling

- After receiving stones or boulders from the quarry, they are crushed into smaller size as required to meet the standard of designer.
- Crushed material is then sieved and different size are separated.
- The crushers are also classified based on the stage of crushing as below:

Primary Crusher:

- At this stage, stones are directly received from the quarry and first reduction in size takes place
- The aggregates obtained will be coarser and non-uniform.
- Jaw crushers, Revolving crusher and Hammer mills are the equipment used at this stage

Aggregate Production and Handling

Secondary Crusher:

- The crushed stones received from the first stage are put into the secondary crusher to further reduce the size of the aggregates.
- Concrete crushers, Roll crushers and Hammer mills are the equipment used at this stage

Tertiary Crusher:

- Final stage after which the aggregates of required sizes are obtained.
- However some stones need to pass through further more crushers also to reduce the size.
- Roll crushers, Roll mills, and Ball mills are the equipment used for this stage.

Aggregate Production and Handling

Screening:

- It is done to eliminate or separate over size crushed stones.
- Screen cloth, revolving screen and vibrating screens are used for screening of aggregates.
- Washing of aggregates is also essential task.
- For mass production, aggregates are screened from multideck screening plant
- Different transporting equipment (e. g. truck, dumpers, conveyor belt etc.) are used.
- Capacity of screen is given by formula, **Q = ACEDG**

Where,

Q = capacity of screen (ton per hour)

A = Area of screen (Square feet)

C = Theoretical capacity of screen (ton per hour/ square feet)

E = Efficiency factor

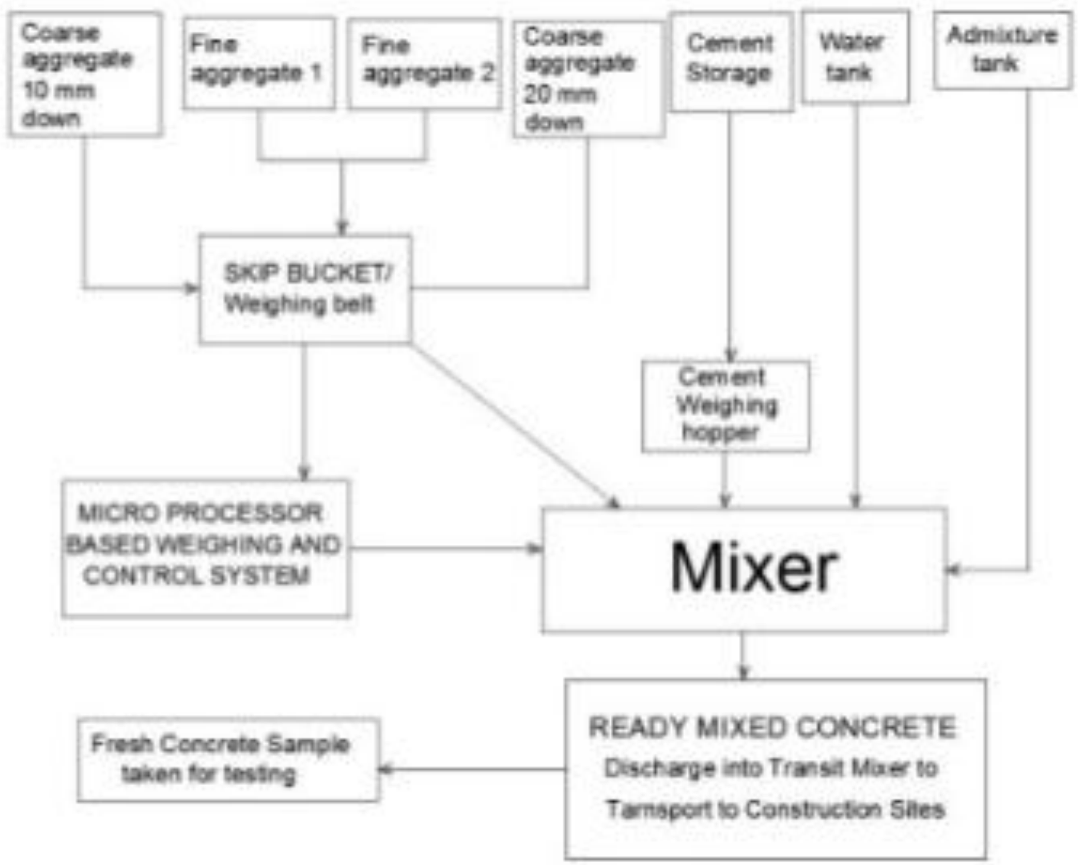
D = Deck Factor

G = Aggregate size factor

Concrete Construction

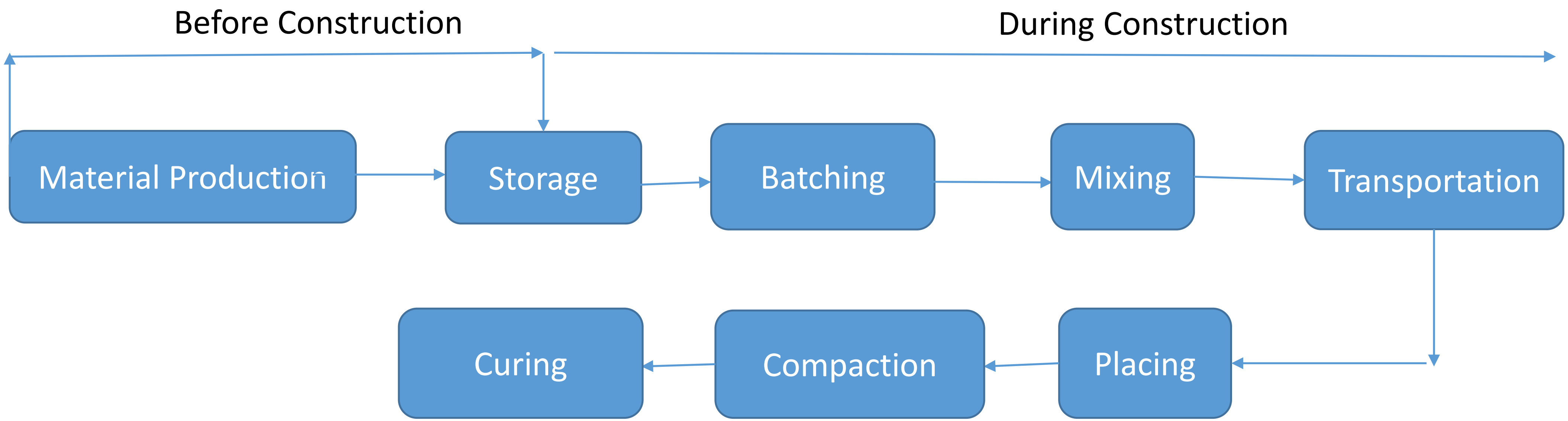
- Concrete production is a stage of concrete construction

Manufacturing Process of Ready-mix Concrete



Concrete Construction

- The following arrow diagram makes us clear about the concrete construction procedure



Concrete Construction

Martial Production:

- Cement: Direct from Factory/ Shop
- Sand: From River
- Coarse Aggregates: From river or From crusher after seiving

Storage:

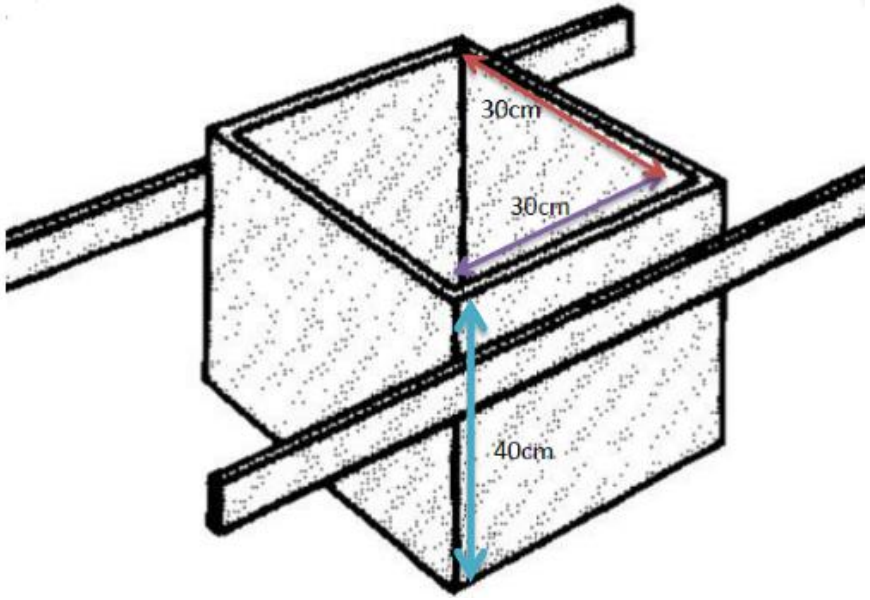
- Cement requires special storage because it is very sensitive to the moisture proof platform
- In huge construction works, they are transported by special means powder cement is stored in cement silo.

Batching:

- Volumetric Batching: Mixing by proportion of Volume. Suitable for small job.
- Weight Batching: Mixing by proportion of weight. Suitable for large construction.

Equipment for Batching:

a) Gauge Box: Used for volumetric batching and small scale work.



Gauge Box

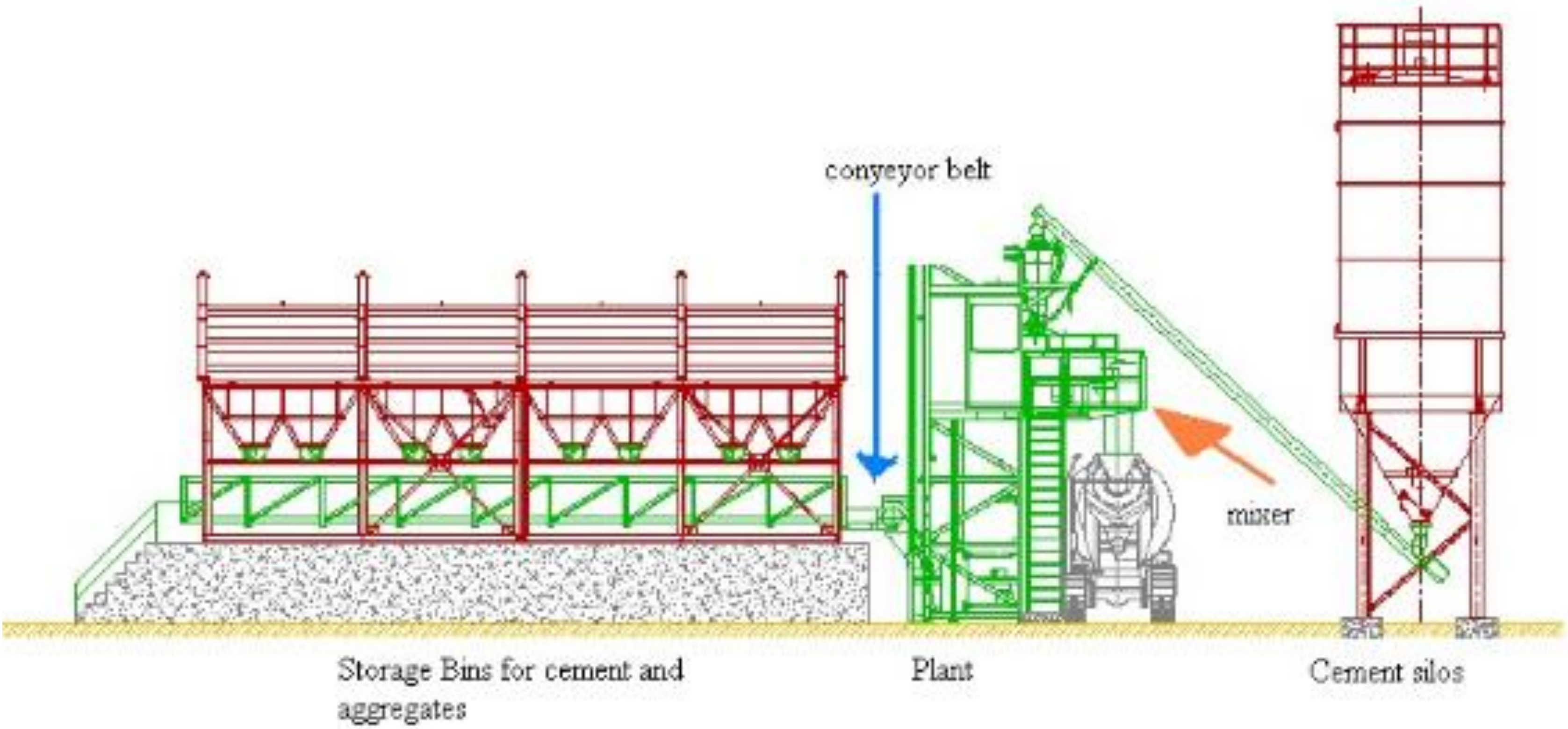
b) Ordinary batching Scale:

- Used for small and medium scale works
- Used for weight batching
- Generally used in lab
- Time consuming and costly

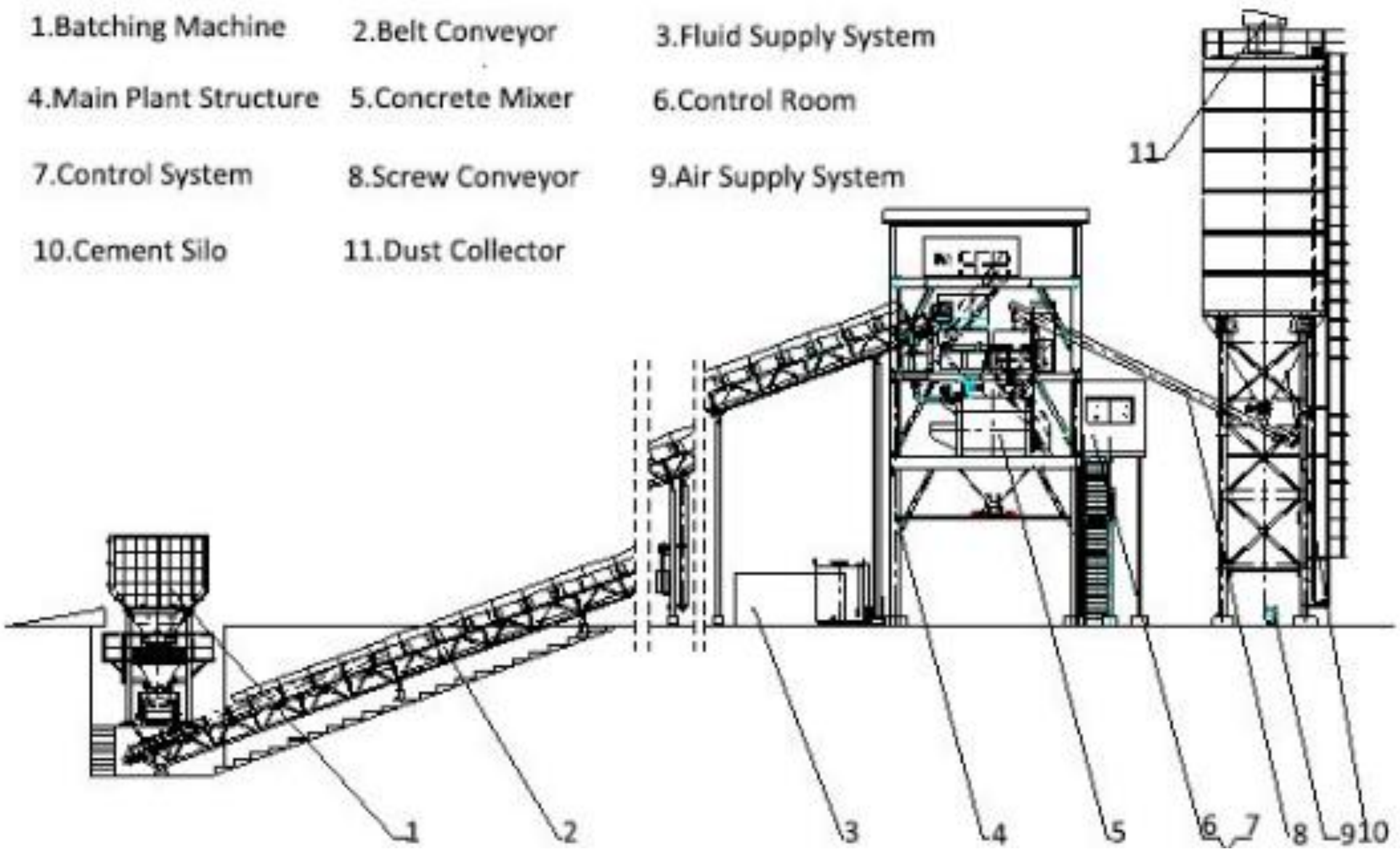
c) Batching Plant:

- Large scale job are done through automatic batching plant
- The main function of it is to receive various ingredient of concrete, weight them to required proportion and then transfer to mixing plant.

Concrete Batching Plant



Concrete Batching Plant



Concrete Mixing

- To produce homogeneous and uniform concrete, mixing of ingredients should be done properly.
- For small scale work ($V < 1$ m3), hand mixing is suitable suitable
- To obtain uniform concrete, turn three times
- For large scale work ($V > 1$ m3), machine mixing is done
- It is more efficient and economical than hand mixing

Methods of Mixing:

Hand Mixing: Used for small scale work. Add 10 % more cement.

Machine Mixing: For large scale work

Equipment Used for Concrete Mixing

a) **Hand Fed Drum Mixer**
b) **Self Loading Concrete Mixture**
c) **Concrete Mixing Pan**
d) **Transit Mixture**
e) **Mixing Plant**

Concrete Construction

Concrete Mixing Pan

Concrete Transit Mixture

Transportation (Handling & Placing)

- Concrete should be mixed, handled, placed in position and compacted as quickly as possible after the water is added.
- In no cases this time shouldn't exceed initial setting time of cement.
- For late setting of concrete, admixtures can be added.
- During handling, improper method may lead to the segregation of concrete which lead to low strength of it.
- The most common equipment for handling of concrete are as below:

Metal Pan, Wheel barrows, Builders hoist, Mini dumper, Conveyer belt system and Concrete pumps.

Compaction of Concrete: It is the process of obtaining the dense concrete mass by removing the entrapped air. Poor compaction leads to honey comb concrete which reduces the strength of concrete. So, optimum compaction is needed.

Equipment: Needle Vibrators, Surface Vibrator, Form Vibrator and Tamping rod

Concrete and Curing

Curing of Concrete:

- Hydration of cement is the cause of hardness and strength of concrete mass.
- Hydration takes a long time even for a year
- But maximum of strength is gained in about below 10 days.
- To attain maximum strength, concrete should be cured with adequate moisture.
- The loss of moisture due to evaporation is accomplished by keeping surface wet.

Methods of concrete curing are:

- Sprinkle water at regular interval
- Cover concrete surface with film of chemical/ jute mesh/ layer of sand which prevents from evaporation of concrete surface.

Tunnel Construction

- Tunnel is the underground passage for travelling and navigation.
- It is an artificially constructed underground passage to by- pass obstacles safely without disturbing the over burden.
- Tunneling is desirable when Rapid transport facilities are required which needs to avoid acquisition of land for roads.
- Tunnels are also erected when shortest route connection is needed in cities.
- Tunnels permit easy gradient & encourage high speed on strategic routes.

Purpose:

- Passage way for roads and rail roads
- Conduits for water supply/ hydropower
- Access to mines and underground space
- Passage for Pedestrians etc.
- The primary objectives of the tunnel alignments are to reduce transit trip times, increase quality and reliability of service and minimize impacts of surface transit operations in sensitive locations.

Tunnel Construction

- Ratio of length to width, in a tunnel , should always be at least in 2: 1.
- A tunnel must be completely enclosed on all sides along the length.

The types of tunnels are classified based on three aspects:

- Based on purpose (road, rail, utilities),
- Based on surrounding material (soft clay vs. hard rock) &
- Submerged tunnels

Economics of Tunneling depend on:

- Nature of Soil/ rock
- Requirements of fill
- Depth of cut > 18m –tunneling

Tunnel Construction

Tunnel Design Criteria:

- Tunneling requires proper design.
- Every tunnel will have its own geometry, design, alignment, and construction methods.

Tunnel design criteria include the following:

- Spatial Requirements;
- Alignment;
- Underground Stations;
- Fire Life Safety; and
- Tunnel Systems and Operation.
- Every tunnel will have its own horizontal and vertical alignment, Tunnel Ventilation, Tunnel Lighting, Electrical and Safety Equipment, Drainage, Fire Life Safety, Security.

Tunnel Construction

The following factors should be taken into consideration when selecting the method of Tunnel construction:

- Tunnel dimensions
- Tunnel geometry
- Length of tunnel
- Total volume to be excavated
- Geological and rock mechanical conditions
- Ground water level and expected water inflow
- Vibration restrictions &
- Allowed ground settlements

Tunnel Construction

Selection of Tunnel alignment depends on:

- Topography of area & points of entrance and exit
- Selection of site of tunnel to be made considering two points
- Alignment restraints
- Environmental Considerations
- Tunnel portal – the interface point of the open cut and the cut and cover tunnel

Classification of Tunnels:

Based on Alignment: Off-Spur tunnels, Short length tunnels to negotiate minor obstacles.

Saddle or Base Tunnels : Tunnels constructed in valleys along natural slope.

Slope Tunnels : Constructed in steep hills for economic and safe operation.

Spiral Tunnels : constructed in narrow valleys in form of loops in interior of mountains so as to increase length of tunnel to avoid steep slopes.

Tunnel Construction

Conventional method , Tunnel Boring Machine (TBM Method) and New Austrian Tunneling Method (NATM)

Conventional Method of Tunnel Construction (Steps):

- Excavation of earth, Removal of excavated earth and Lining and strengthen of wall and ceiling

Conventional Method of Tunnel Construction (Process):

- Setting up & drilling
- Loading holes and softening explosives
- Ventilating and removing the dust following an explosion
- Loading and hauling muck
- Removing ground water if necessary
- Erecting supports for roof and sides if necessary
- Placing reinforcing steel
- Placing the concrete lining

Equipment Used in Tunnel Construction

- Drilling machine
- Electric detonating arrangement
- Ventilating arrangement
- Wheel barrow, trucks and loader, rail buggies (muck cars)
- Pumps
- Formwork and support
- Concreting machine/ batching plant/ concrete pump

Introduction to TBM

Introduction

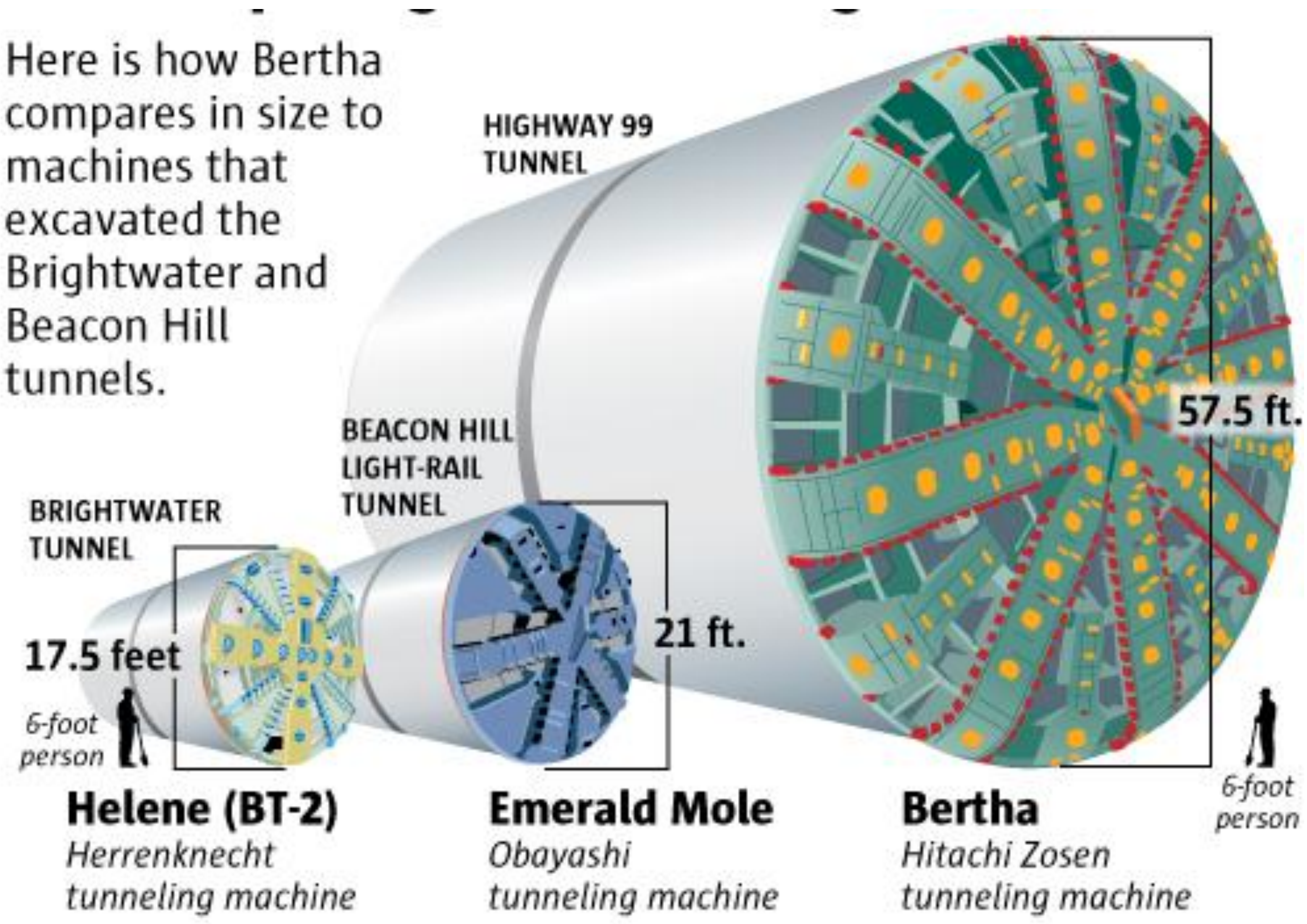
- ➢Tunnel Boring Machine (TBM) is used to excavate tunnels with a circular cross section.
- ➢They can be used on any geologies.
- ➢ It can bore tunnel of diameter 1 to 20 meter
- ➢ It made tunnel boring process risk fee.

Drawback Using TBM

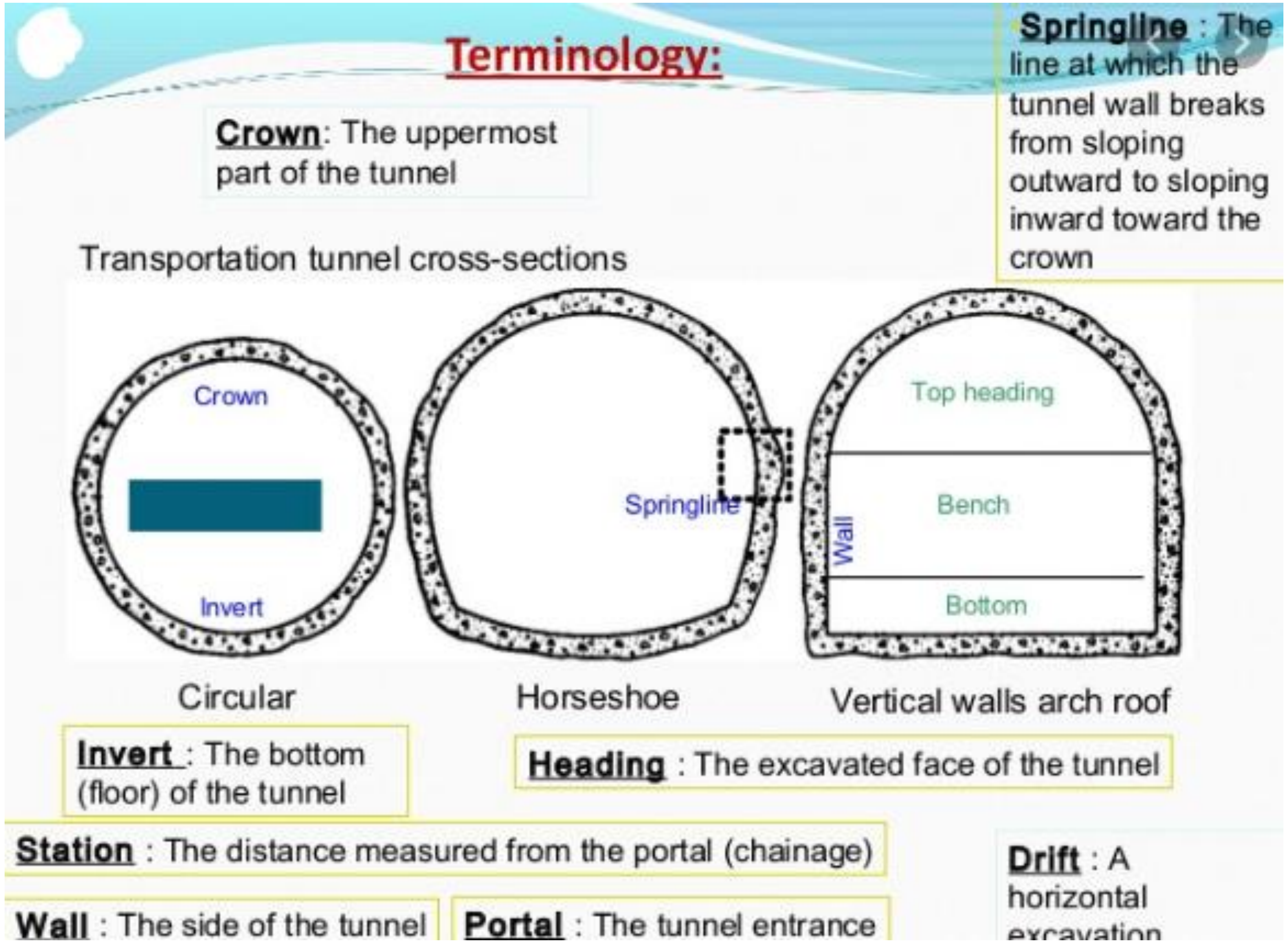
- ➢High cost due to the need of supporting things such as conveyer belt, rails, slurry separating plant, slurry pipelines etc.
- ➢High cost of associates supplies e. g. drill bits, TBM cutters, blasting agents etc.

Tunnel Boring Machine (TBM)

Comparison of Different Types of TBM



Terminology Related to Tunnel Boring



4.3 Selection of Appropriate Equipment

Importance of Construction Equipment

- In Case of major construction projects, the speed of work and the timely completion of work is very important.
- The mechanization of most of the construction work is required, in which the construction equipment play the most important role.
- The proper use of the appropriate equipment contribute to economy, quality, safety, speedy and timely completion of the project.
- The cost of construction is a major factor in all the projects.
- The factors that influence construction costs mainly are materials, labor, construction equipment, overhead and profit.

Importance of Construction Equipment

- The cost of construction equipment for civil engineering construction projects ranges from 25 % to 40 % of the total project.
- The amount which is invested in the purchase of a construction equipment should be recovered during the useful period of such equipment.
- The construction equipment are deployed on the construction projects for the various reasons, such as

Criteria for Selecting Construction Equipment

Technology:

- The focus should be on machines that are appropriate for the job at hand, the environment and terrain it needs to work in and the people operating the machine.
- Technical literacy amongst operators is a rather difficult thing to come by in operators these days.
- As a matter of fact, most operators find these advanced machines to be more of a hassle.
- The make of the important parts like engine, hydraulics and the transmission etc. play a vital role in the service premium of the machine and its resale value.
- So look for a machine where technology is appropriate.

Criteria for Selecting Construction Equipment

Fuel Efficiency:

- Fuel usually consists up to 60% of the costs involved and so fuel efficiency should be given prime importance.
- Experts recommend that fuel consumption be measured and considered based on cost per unit of output instead of measuring it by the hour.
- Double and triple check how efficient the machine in question is before buying or renting it.

After Sales support:

- Expertise in matter of heavy construction equipment is a difficult thing to find.
- So, when choosing a machine ensure that the replacement parts are easily available and appropriately priced.
- Additionally, ensure that adequate after sales support is available at short notices since machine down time is usually not welcome.
- Ensure that there is adequate appropriate support available before zeroing in onto machine.

Criteria for Selecting Construction Equipment

- **Reliability and ruggedness:**
- The machines should be suited to be able to handle the vast differences of the terrain and climate that the various job sites.
- Machines have to reliable and asking for more than 95% uptime is not stretching it given the kind of technology available today.
- An unreliable machine will have both direct and collateral cost damages.
- Say if a asphalt compact was to break down, it would leave a truck full of hot bitumen stranded, adding to the cost of the project.
- So, reliable rugged machines should be preferred.

Criteria for Selecting Construction Equipment

Price:

- Ensure that the machine is appropriately priced for what it has to offer.
- Considering a 10% per year depreciation value, a machine should last for atlas 10 years, pay for itself and earn profits if one has to buy a machine.
- Most buyers ask cheaper products and expect them to last at least 2-4 project cycles.
- When it comes to machines various costs come into play.
- The yearly depreciation of the machine’s value, fuel costs, insurance, storage, maintenance, logistic costs, operator costs and spare parts cost to name a few.
- Take these things into consideration when wanting to select the construction equipment.